Paleomagnetism of Lava Flows from Kauai, Hawaii

RICHARD R. DOELL[1]

U.S. Geological Survey, Menlo Park, California 94025

A paleomagnetic study of 571 sample cores drilled from 120 lava flows on the island of Kauai yields low values of paleosecular variation. These data show that the low secular variation reported for recent times and for the Brunhes normal polarity epoch was a distinctive feature of the central Pacific Ocean at least as long ago as 5 m.y. The mean virtual geomagnetic pole is displaced 6.4° on the other side of the geographic pole from Kauai; this displacement lends some support to the hypothesis of a 'main' axial geomagnetic dipole displaced away from the center toward the north pole.

Studies of geomagnetic field behavior in the Hawaiian Islands are of primary interest in terms of secular variation because of the present anomalously low secular variation in the central Pacific Ocean [*Fisk*, 1931] and the missing nondipole field there [*Cox*, 1962; *Creer*, 1962; *Doell and Cox*, 1971]. The permanence of this feature through the Brunhes polarity epoch has been established in previous paleomagnetic studies [*Doell and Cox*, 1963, 1965; *Doell*, 1969, 1972]. One of the primary purposes of the present investigation is to pursue the matter of the permanence of this feature back to 5 m.y. ago, the age of the oldest rocks on Kauai [*McDougall*, 1964].

The only other paleomagnetic study of rocks from Kauai was that made by *Tarling* [1965] on 12 samples from 4 sites. From that small collection no conclusions about paleosecular variation were possible.

GEOLOGICAL SETTING

The geology of the island of Kauai has been described by *Macdonald et al.* [1960], and the following summary is taken from their work. Kauai consists mainly of one large shield volcano made up of the Waimea Canyon volcanic series, which is divided into the following formations: (1) the Napali formation, which forms the major part of the shield and consists of thin basalts dipping away from the summit; (2) the Olokele formation, a thick series of ponded lavas that later filled a large caldera at the summit; (3) the Haupu formation, a series of ponded lavas that filled a small caldera to the southeast; and (4) the Makaweli formation, a similar series of ponded lavas that filled a graben southwest of the caldera. After a long period of quiescence, the Koloa volcanic series was erupted from a large number of vents over the eastern half of the island.

McDougall [1964] has obtained potassium-argon dates of 4.5–5.6 m.y. and 3.5–4.0 m.y. on samples from the Napali and Makaweli formations, respectively He also obtained a date of 1.4 m.y. on one sample from the Koloa volcanic series. These age data are compatible with the Pliocene age assignment given by *Macdonald et al.* [1960].

A very generalized geologic map, Figure 1, shows the location of sampling sites for the paleomagnetic investigations reported here. Lavas of the Napali formation were sampled in four locations (see appendix for details): low on the north (lavas K1–K3) and southwest (K19–K60) edges of the island, high on the west rim of Waimea Canyon (K7-K18), and at the bottom of the canyon (K4–K6). *McDougall* [1964] obtained two samples for age measurements from the same area on the west rim of Waimea Canyon and one from lower exposures to the southwest. Lavas of the Makaweli formation were collected from one locality in the bottom of Waimea Canyon (K61–K84). McDougall's dating samples came from sites farther downstream to the south and at higher elevations to the southeast. Several flows of the Koloa volcanic series were sampled at scattered localities as shown in Figure 1. Most Koloa samples, however, were obtained from a superposed se-

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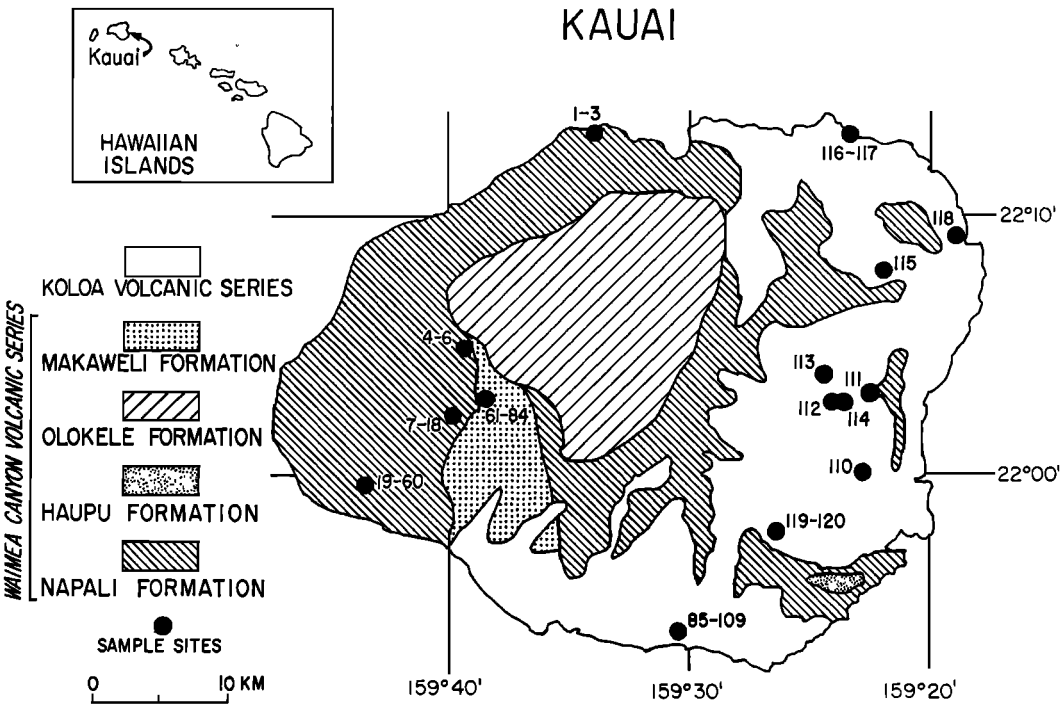


Fig. 1. Generalized geologic map of Kauai [after *Stearns*, 1966, p. 199], showing location of sampling sites for paleomagnetic studies. Sample site numbers correspond to lava flow numbers in Table 1.

quence (K85–K109) in Lawai Gulch, the location from which McDougall obtained the dating sample mentioned above. The Olokele and Haupu formations were not sampled.

METHODS AND RESULTS

All samples collected for the present investigation were drilled and geographically oriented with portable equipment; remanent magnetization (RM) was measured with spinner type magnetometers; and testing of stability of RM and magnetic cleaning were performed by means of partial alternating-field (af) demagnetization.

Eight samples were drilled from each lava flow in the Makaweli formation, and five from most of the lavas in the Napali formation and Koloa volcanic series. For many lavas of the latter two geologic units, however, only one or two cores were drilled. One specimen from each core collected at a given site was measured. Flow-average palemagnetic data are given in Table 1.

Various amounts of af cleaning were needed to remove anomalous remanence (Table 1, column 4), and a few specimens with widely divergent directions were excluded from analysis (Table 1, column 2). The rationale used in selecting a particular alternating field for magnetic cleaning (which included progressive af demagnetization up to at least 400-oe peak field strength for each lava reported here) and for excluding specimens from flow-average analysis are considered in detail in another paper [*Doell*, 1970].

To maintain statistical uniformity [*Doell and Cox*, 1972], the paleomagnetic data from lava flows represented by fewer than four samples or with α_{95} confidence values of less than 9.0° (units marked with an asterik in Table 1) have not been included in the over-all paleomagnetic analysis. Most of these lava flows showed evidence of stability under af demagnetization, so that the magnetic polarities as recorded for them in Table 1 are probably reliable. For a few flows (marked by a dagger in Table 1), the progressive af demagnetization studies did not indi-

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TABLE 1. Flow-Average Data

Unit	No. of Samples‡	Maximum Lateral Distance, meters	a.f. Cleaning (Peak oe)	RM Decl., °E	RM Incl., deg dn	Length of Resultant R	Precision Parameter k_e	95% Confidence, deg	Angular s.d., deg	VGP Lat., °N	VGP Long., °E
Miscellaneous Collections in the Koloa Volcanic Series											
K120	4 (4)	35	100	171.0	−38.5	3.9946	554	3.9	3.4	−81.6	290.2
119	4 (5)	32	NRM	173.4	−31.5	4.9856	278	4.6	4.9	−82.1	253.2
118*†	2 (2)	6	100	295.4	−71.5				3.2	−5.3	50.9
117*	1 (1)		NRM	166.0	−32.8					−76.2	274.8
116*	2 (2)	8	NRM	2.0	32.4				6.2	85.0	357.9
115*	2 (2)	1	200	197.8	−30.9				2.0	−72.4	125.7
114*	2 (2)	8	100	18.2	7.6				8.2	64.6	333.9
113*	(2)	183	(no agreement in directions)								
112*	1 (1)		NRM	359.9	28.4					83.1	21.4
111*	2 (2)	5	NRM	2.5	32.2				3.6	84.8	352.9
110*†	2 (2)	6	100	22.4	9.6				3.2	62.4	325.8
Lawai Gulch Section of Koloa Volcanic Series											
109	4 (5)	11	100	0.6	52.6	3.9959	734	3.4	3.0	78.7	203.0
108	5 (5)	28	200	180.4	−28.2	4.9978	1837	1.8	1.9	−83.1	197.7
107*	3 (3)	16	50	351.7	58.3	2.9927	273	7.5	4.9	71.4	179.9
106*†	2 (2)	2	50	17.7	67.0				5.6	68.1	240.8
105*†	3 (3)	3	100	16.6	53.0	2.9773	88	13.2	8.6	71.3	248.4
104	5 (5)	7	100	15.1	9.0	4.9475	76	8.8	9.3	67.3	338.3
103*†	2 (2)	2	600	88.0	−80.9				13.4	−20.2	1.6
102*	2 (2)	3	200	171.4	−36.0				7.2	−81.5	274.2
101*	1 (1)		200	168.0	−24.2					−75.4	253.6
100*	2 (2)	1	200	173.2	−29.5				3.9	−81.2	248.3
99*	2 (2)	4	NRM	176.0	−30.2				2.9	−83.2	235.1
98	5 (5)	5	NRM	179.8	−34.1	4.9817	219	5.2	5.5	−86.8	203.8
97	5 (5)	12	NRM	178.2	−31.4	4.9991	4287	1.2	1.2	−84.8	219.6
96	5 (5)	13	NRM	180.6	−30.1	4.9932	585	3.2	3.4	−84.2	194.4
95	5 (5)	4	NRM	181.0	−30.2	4.9977	1729	1.8	2.0	−84.2	190.6
94	5 (5)	17	NRM	175.6	−33.5	4.9971	1366	2.1	2.2	−84.5	250.6
93	5 (5)	8	NRM	171.5	−27.3	4.9839	249	4.9	5.1	−79.0	249.4
92	5 (5)	16	NRM	174.1	−29.6	4.9976	1689	1.9	2.0	−81.8	244.2
91	5 (5)	26	NRM	175.8	−34.6	4.9916	474	3.5	3.7	−85.2	255.1

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TABLE 1. (*Continued*)

Unit	No. of Samples‡	Maximum Lateral Distance, meters	a.f. Cleaning (Peak oe)	RM Decl., °E	RM Incl., deg dn	Length of Resultant R	Precision Parameter k_c	95% Confidence, deg	Angular s.d., deg	VGP Lat., °N	VGP Long., °E
90	5 (5)	9	NRM	177.6	−32.6	4.9954	877	2.6	2.7	−85.2	229.6
89	5 (5)	15	NRM	179.0	−32.6	4.9977	1757	1.8	1.9	−85.7	212.7
88	5 (5)	12	200	174.4	−29.2	4.9857	281	4.6	4.8	−81.8	241.8
87	5 (5)	8	NRM	177.7	−32.4	4.9969	1293	2.1	2.2	−85.2	227.4
86	5 (5)	46	NRM	177.8	−31.6	4.9858	281	4.6	4.8	−84.8	224.2
85	5 (5)	12	NRM	178.6	−21.9	4.9843	255	4.8	5.1	−79.4	207.9
				Waimea Canyon Section of Makaweli Formation							
84	8 (8)	9	50	192.8	−37.6	7.9871	544	2.4	3.5	−78.1	112.8
83	8 (8)	12	25	196.0	−31.2	7.9825	401	2.8	4.0	−74.1	126.6
82	8 (8)	17	50	194.2	−35.1	7.9883	598	2.3	3.3	−76.4	119.1
81	8 (8)	13	100	192.8	−30.8	7.9810	368	2.9	4.2	−76.8	132.6
80	8 (8)	13	NRM	196.7	−28.8	7.9895	667	2.2	3.1	−72.9	130.4
79	8 (8)	9	NRM	190.6	−28.8	7.9828	407	2.8	4.0	−78.0	142.3
78	9 (9)	9	NRM	194.7	−29.1	8.9871	620	2.1	3.3	−74.6	132.9
77	8 (8)	11	NRM	192.1	−31.0	7.9824	398	2.8	4.1	−77.4	133.4
76	8 (8)	16	50	194.1	−28.4	7.9920	881	1.9	2.7	−75.0	135.5
75	8 (8)	16	NRM	191.5	−27.6	7.9719	249	3.5	5.1	−76.8	142.6
74	9 (9)	24	NRM	195.7	−26.5	8.9776	357	2.7	4.3	−73.1	136.1
73	8 (8)	20	50	191.2	−27.3	7.9742	272	3.4	4.9	−77.0	144.1
72	9 (8)	11	200	184.5	−26.5	7.9112	79	6.3	9.1	−80.8	171.5
71	8 (8)	13	50	178.8	−32.5	7.9840	439	2.6	3.9	−85.5	215.4
70	9 (9)	24	100	173.0	−24.9	8.9879	663	2.0	3.2	−78.8	238.2
69	8 (8)	8	NRM	178.1	−23.2	7.9941	1186	1.6	2.4	−80.0	210.9
68	9 (9)	14	50	176.2	−24.2	8.9891	734	1.9	3.0	−79.9	221.8
67	9 (9)	17	100	178.7	−20.7	8.9818	439	2.5	3.9	−78.6	206.7
66	9 (9)	15	100	176.7	−20.8	8.9816	436	2.5	3.9	−78.2	216.3
65	8 (8)	8	50	176.9	−19.6	7.9892	650	2.2	3.2	−77.6	215.0
64	8 (8)	28	100	184.6	−38.4	7.9366	110	5.3	7.7	−85.7	115.5
63	8 (8)	36	50	193.3	−46.0	7.9897	679	2.1	3.1	−76.8	83.9
62	7 (7)	18	200	183.0	−26.0	6.9881	502	2.7	3.6	−81.2	181.0
61	8 (8)	14	NRM	181.3	−25.5	7.9880	582	2.3	3.4	−81.2	191.9

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TABLE 1. (*Continued*)

Unit	No. of Samples‡	Maximum Lateral Distance, meters	a.f. Cleaning (Peak oe)	RM Decl., °E	RM Incl., deg dn	Length of Resultant R	Precision Parameter k_e	95% Confidence, deg	Angular s.d., deg	VGP Lat., °N	VGP Long., °E
					Puehu Ridge Section of Napali Formation						
K 60	5 (5)	6	100	359.6	29.0	4.9917	484	3.5	3.7	83.4	24.1
59	5 (5)	5	100	359.9	36.0	4.9947	751	2.8	3.0	87.9	22.6
58	5 (5)	6	100	5.7	13.9	4.9919	496	3.4	3.6	74.0	359.2
57	5 (5)	10	NRM	7.5	16.6	4.9914	464	3.6	3.8	74.6	351.2
56	5 (5)	15	100	7.2	17.3	4.9885	349	4.1	4.3	75.2	351.4
55	5 (5)	7	NRM	0.6	7.8	4.9854	273	4.6	4.9	71.9	18.5
54	5 (5)	7	NRM	1.8	10.7	4.9705	136	6.6	7.0	73.2	14.0
53	5 (5)	12	NRM	9.2	13.4	4.9916	479	3.5	3.7	72.4	348.6
52	5 (5)	7	NMR	6.3	17.0	4.9979	1934	1.7	1.8	75.4	354.8
51	5 (5)	10	NRM	358.3	12.3	4.9983	2384	1.6	1.7	74.1	26.5
50	5 (5)	7	NRM	356.5	11.6	4.9911	450	3.6	3.8	73.4	32.6
49	5 (5)	18	NRM	0.4	16.4	4.9962	1041	2.4	2.5	76.4	18.6
48	5 (5)	6	50	356.4	12.5	4.9906	424	3.7	3.9	73.9	33.1
47	5 (5)	7	50	356.5	13.3	4.9917	482	3.5	3.7	74.3	33.2
46	5 (5)	17	NRM	1.7	12.6	4.9894	378	3.9	4.2	74.3	14.1
45	5 (5)	5	NRM	3.0	36.0	4.9951	818	2.7	2.8	86.5	327.0
44	5 (5)	6	NRM	4.5	36.7	4.9972	1455	2.0	2.1	85.5	310.9
43	5 (5)	3	NRM	355.7	40.6	4.9936	630	3.0	3.2	85.9	127.8
42	5 (5)	7	NRM	354.8	40.0	4.9930	575	3.2	3.4	85.2	119.6
41	7 (7)	6	NRM	354.4	42.7	6.9884	519	2.6	3.6	84.2	139.3
40	5 (5)	3	NRM	354.8	48.2	4.9886	352	4.1	4.3	81.4	167.9
39	5 (5)	24	NRM	351.9	50.0	4.9877	325	4.2	4.5	78.6	162.5
38	5 (5)	7	NRM	356.7	51.1	4.9908	436	3.7	3.9	79.8	184.2
37	5 (5)	5	NRM	358.8	47.4	4.9850	266	4.7	5.0	83.5	191.3
36	5 (5)	3	50	0.6	46.2	4.9960	1010	2.4	2.6	84.5	206.3
35*	4 (5)	3	100	353.5	26.4	3.9690	97	9.4	8.2	79.8	58.8
34	5 (5)	6	NRM	346.9	29.1	4.9877	324	4.3	4.5	76.0	84.8
33	5 (5)	8	NRM	346.8	31.5	4.9931	579	3.2	3.4	76.6	90.7
32	5 (5)	9	NRM	352.9	34.9	4.9898	391	3.9	4.1	82.8	89.0
31	5 (5)	4	NRM	350.4	33.2	4.9987	3187	1.4	1.4	80.2	88.5
30	5 (5)	11	NRM	351.0	34.7	4.9895	380	3.9	4.2	81.0	92.5
29	5 (5)	7	NRM	3.8	27.5	4.9940	665	3.0	3.1	81.7	353.9
28	5 (5)	10	NRM	6.4	29.5	4.9950	809	2.7	2.8	81.3	335.1
27	5 (5)	4	NRM	5.7	26.8	4.9942	684	2.9	3.1	80.5	344.6

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TABLE 1. (*Continued*)

Unit	No. of Samples‡	Maximum Lateral Distance, meters	a.f. Cleaning (Peak oe)	RM Decl., °E	RM Incl., deg dn	Length of Resultant R	Precision Parameter k_e	95% Confidence, deg	Angular s.d., deg	VGP Lat., °N	VGP Long., °E
26	5 (5)	3	NRM	341.7	13.9	4.9935	614	3.1	3.3	66.8	72.8
25	5 (5)	11	NRM	344.8	12.4	4.9908	436	3.7	3.9	68.5	65.5
24	5 (5)	5	NRM	343.5	9.1	4.9948	762	2.8	2.9	66.3	65.0
23	5 (5)	4	100	331.7	17.2	4.9928	554	3.2	3.4	59.8	88.8
22	5 (5)	10	200	334.1	20.9	4.9972	1419	2.0	2.2	62.8	90.1
21*	1 (1)		100	352.2	42.4					82.4	131.4
20	5 (5)	9	NRM	351.7	36.6	4.9877	325	4.2	4.5	82.1	99.7
19	5 (5)	5	NRM	351.7	38.2	4.9956	915	2.5	2.7	82.3	107.4
				Waimea Canyon Section of Napali Formation							
18*	2 (2)	2	NRM	358.0	41.5				3.9	87.4	155.9
17*	1 (1)		200	2.2	39.0					88.0	290.2
16*	1 (1)		200	9.1	46.0					80.2	255.5
15*	1 (1)		200	328.9	12.6					56.1	87.5
14*	1 (1)		200	331.7	15.5					59.3	87.3
13	5 (5)	10	200	338.2	8.1	4.9775	178	5.8	6.1	62.2	73.2
12*	1 (1)		200	354.2	24.5					79.3	52.2
11*	1 (1)		200	319.9	23.5					50.6	102.8
10	5 (5)	8	100	356.0	13.6	4.9755	163	6.0	6.3	74.4	35.4
9	5 (5)	6	NRM	6.1	34.0	4.9852	271	4.6	4.9	83.3	320.2
8*	1 (1)		200	3.0	46.4					83.7	225.4
7*	1 (1)		200	359.7	51.3					80.1	198.9
6	4 (4)	6	NRM	183.5	−31.3	3.9940	496	4.1	3.6	−83.9	166.9
5	4 (4)	11	NRM	185.0	−32.4	3.9986	2185	2.0	1.7	−83.5	153.5
4	4 (4)	10	NRM	184.2	−32.5	3.9948	574	3.8	3.4	−84.0	157.4
				Napali Formation at Dry Cave							
3*	1 (1)		200	359.7	21.8					79.1	22.0
2*	1 (1)		200	2.8	41.2					87.1	261.1
1*	1 (1)		200	3.3	35.2					85.8	332.0

* Not used in paleomagnetic and variance analyses.
† RM unstable to af demagnetization. Directions considered unreliable.
‡ The first number is the number of samples collected; number in parentheses is the number of samples used in analysis.

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cate complete stability for the natural remanent magnetization; the data from these flows therefore could be unreliable.

The over-all stability of the natural remanent magnetization of the lavas sampled on Kauai is summarized in Figure 2 in the form of a histogram of the intensity of RM remaining after progressive partial af demagnetization treatments. Data for one specimen per lava flow are used in this figure. Typically, values of NRM intensity are near 4.0×10^{-3} emu/cm^3; the RM decreases to about 20–25% of this value after treatment at 400-oe peak af strength. This behavior is typical of individual Kauai lava flows and of TRM in Hawaiian lavas in general [*Doell and Cox*, 1963] and suggests that the NRM of the Kauai lavas was acquired when the lavas cooled (possible exceptions are marked with a dagger in Table 1).

The declination and inclination data cited in Table 1 therefore may be taken as those of the geomagnetic field at the time and place of cooling of the individual lavas. The precision with which the directions that have been determined on analysis of data from four or more sample cores per lava is given by the values of the precision parameter k and the 95% confidence semiangle, α_{95} [*Fisher*, 1953], and by the angular standard deviation. The over-all precision is summarized in Figure 3 as a histogram of the α_{95} values; the majority of the directions are defined within 5° at the 95% confidence level, which is sufficiently smaller than the typical flow-to-flow angular variations that the within-flow angular variance does not contribute excessively to the over-all variance.

DISCUSSION

The positions of virtual geomagnetic poles (VGP) calculated from the directions of RM cited in Table 1 are shown in Figures 4, 5, and 6 for the Napali formation, Makaweli formation, and Koloa volcanic series, respectively. The means for the VGP and RM directions for these same three volcanic units and other commonly employed paleomagnetic statistical data are given in Table 2, where only data with four or more cores per lava were used in the analyses. Similar data are cited for all lavas combined.

In general, the magnetic polarity results and radiometric age data obtained by *McDougall* [1964] are in agreement with recent summaries of the polarity epoch time scale [*Cox*, 1968]. The reversed polarity measured in the lower part of the Koloa volcanic series sampled at Lawai Gulch (K85–K98), from which McDougall obtained the sample dated at 1.4 m.y., clearly falls into the Matuyama reversed po-

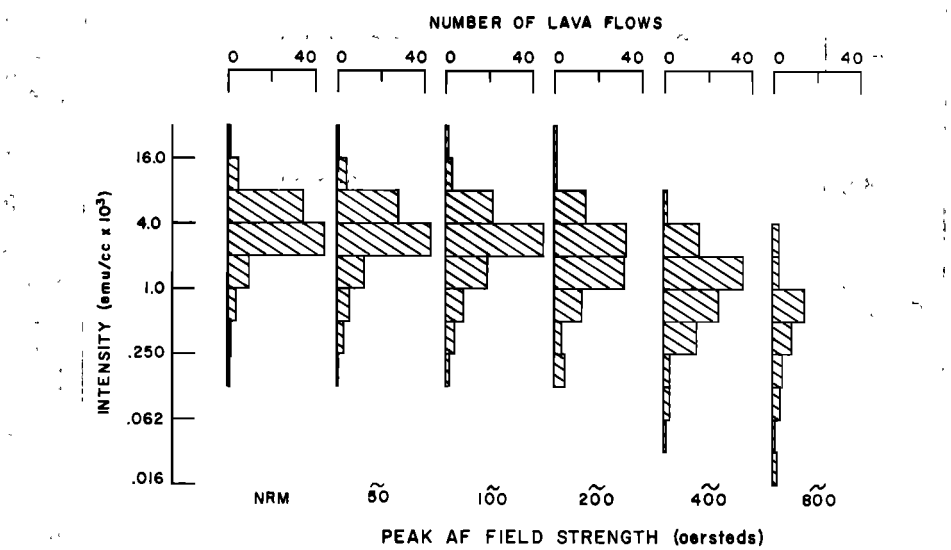


Fig. 2. Histograms of remanent magnetization remaining after partial af demagnetization at peak field strengths of 0 (NRM), 50, 100, 200, 400, and 800 (not all lavas).

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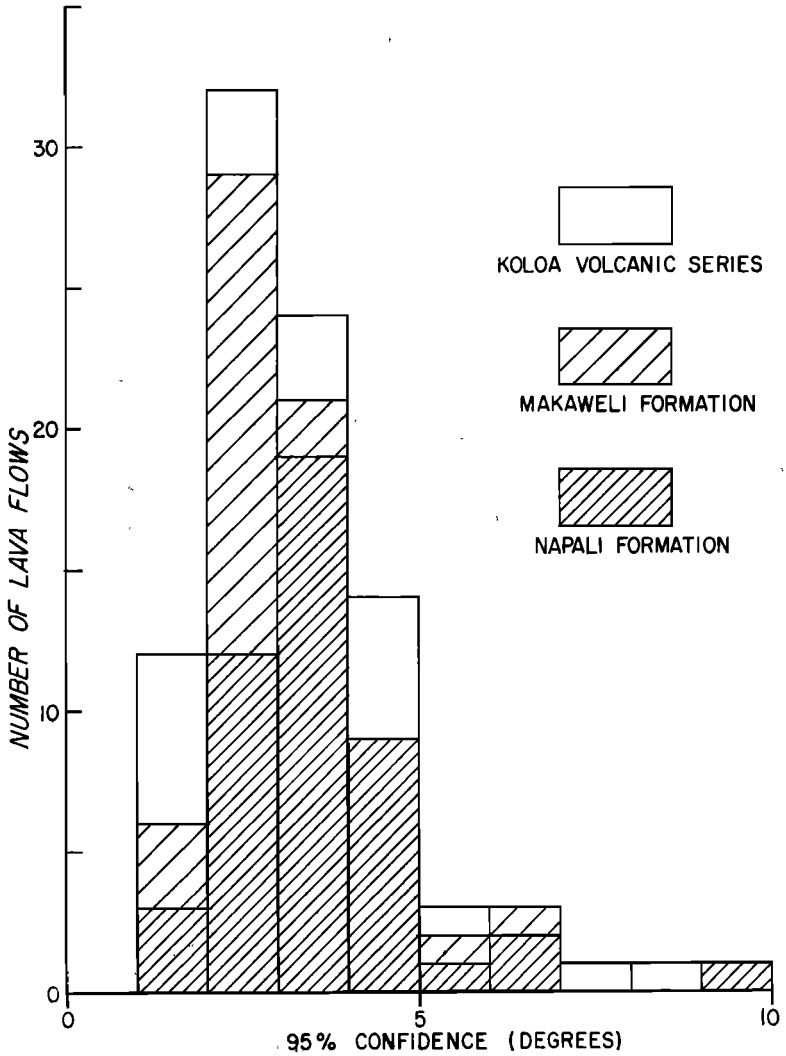


Fig. 3. Histogram of flow-average 95% confidence cone semiangles (α_{95}) for lavas with four or more sample cores.

larity epoch prior to the Jaramillo normal polarity event.

The geologic sequence of the upper flows at Lawai Gulch (K99–K109) is less clear than that of the lower superposed lavas. When samples from these flows were collected, the sequence as given in Table 1, was determined as the most likely history of emplacement. It is quite possible, however, that there are some stratigraphic uncertainties in this sequence. Several of the lavas are quite limited in extent, having been formed in a small stream normal to and near the top of the Lawai exposure. Moreover, it appears that the stream was actively eroding between the times of emplacement of these lavas. Thus, it cannot be stated with certainty whether the normally magnetized lavas K104–K107 were formed during the Jaramillo event or later during the Brunhes normal polarity epoch. The time of formation of lavas K104–K107 would depend on whether or not K108 is younger than K104–K107.

Elsewhere, the Koloa volcanic series is at many places weathered to a lateritic soil. In many exposures, however, it was possible to sample fresh residual boulders ('eggs'), as indi-

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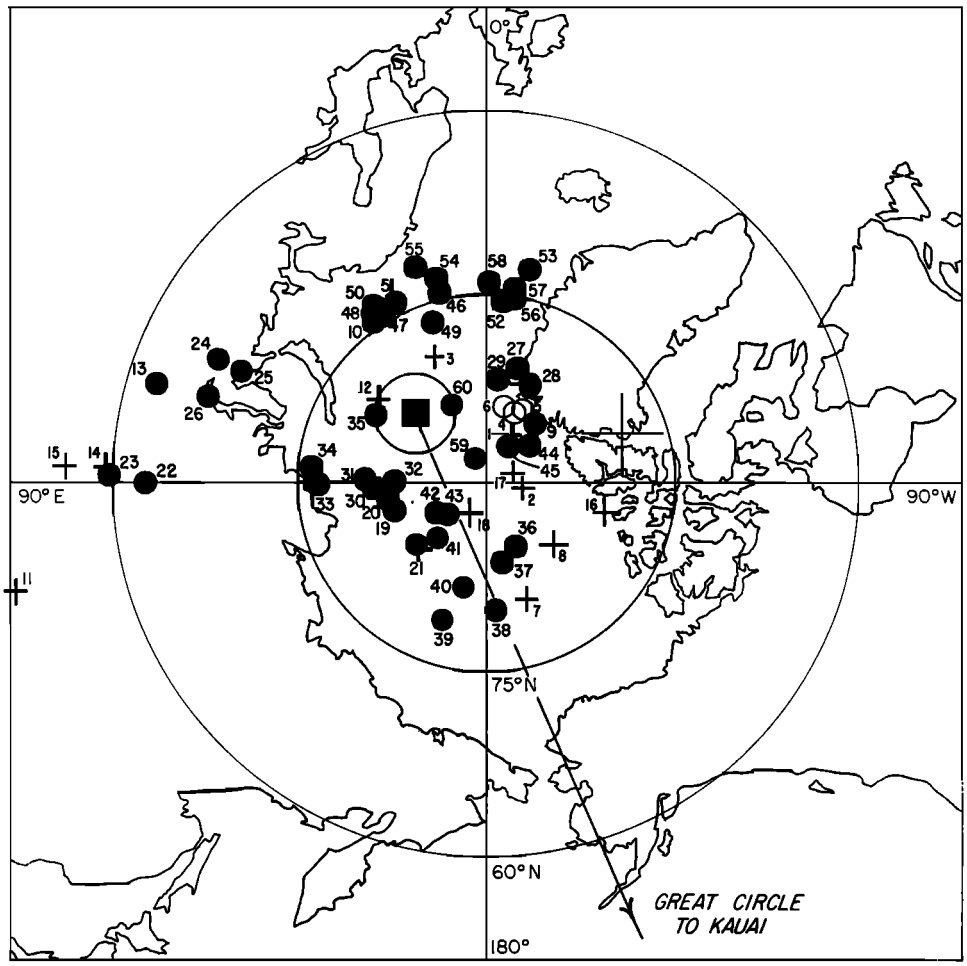


Fig. 4. Virtual geomagnetic pole positions in lavas of the Napali formation. Numbers refer to those of Table 1, column 1. Square, the over-all mean, with the 95% confidence circle drawn about it. Large cross, the position of the geomagnetic pole (which is also the VGP for the present field direction in Kauai). Filled circles (four or more cores per lava) and plus signs (less than four cores per lava), normal polarity; open circles (four or more cores) and crosses on Figure 6 (less than four cores), reversed polarity. Portion of Lambert's equal-area projection.

cated in the appendix. The fact that both reversed and normal polarities were measured in these Koloa volcanic series lavas indicates, as for the Lawai Gulch lavas, that the Koloa was erupted during at least parts of the Matuyama and Brunhes polarity epochs. This finding is in agreement with the suggestion of *Macdonald et al.* [1960] that the Koloa volcanic series was erupted over a long period extending to fairly recent times.

Lavas of the Makaweli formation sampled in this study are all reversely magnetized. Although *McDougall* [1964] did not sample at the same locality, he did obtain samples that bracket, in elevation, the paleomagnetically sampled flows. McDougall argues that these lavas were erupted rapidly near 3.8 to 3.9 m.y. ago; thus they would be placed in the late portion of the Gilbert reversed polarity epoch, in agreement with their observed remanence.

McDougall's [1964] ages for the Napali formation range from about 5.6 m.y. low (elevation, 75 meters on the southeast flank of the volcano to 4.5 m.y. high (elevation, 750 meters)

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on the west rim of Waimea Canyon; he argues that the samples are stratigraphically quite separated and that the separation and the radiometric dates indicate a long duration (1 m.y.) for eruption of the Napali lavas. As the flows of the Napali formation are outward dipping, however, the actual stratigraphic separation need not be much greater than 100 meters.

Paleomagnetic samples from flows K7 through K18 bracket McDougall's samples with ages near 4.5 m.y. and are all normally magnetized. Lava flows K19–K60 are probably stratigraphically lower than K7–K18 but are higher than McDougall's flow dated at about 5.6 m.y. All are normally magnetized. Three Napali lavas sampled deep in Waimea Canyon (K4–K6) are stratigraphically much lower than the 5.6-m.y. lava and are reversely magnetized. The three normal flows (K1–K3) to the north are difficult to relate to the southern lavas.

According to *Cox's* [1968] summary, there are five changes in polarity between 5.6 and 4.5 m.y. ago with a large reversed interval (5.43 to 487 m.y. ago) in the middle of this interval. Thus, it is possible to fit the paleomagnetic and age data into the polarity epoch time scale, but one must wonder why no reversed lavas were encountered in any of the lavas numbered K7–K60 if the flows were

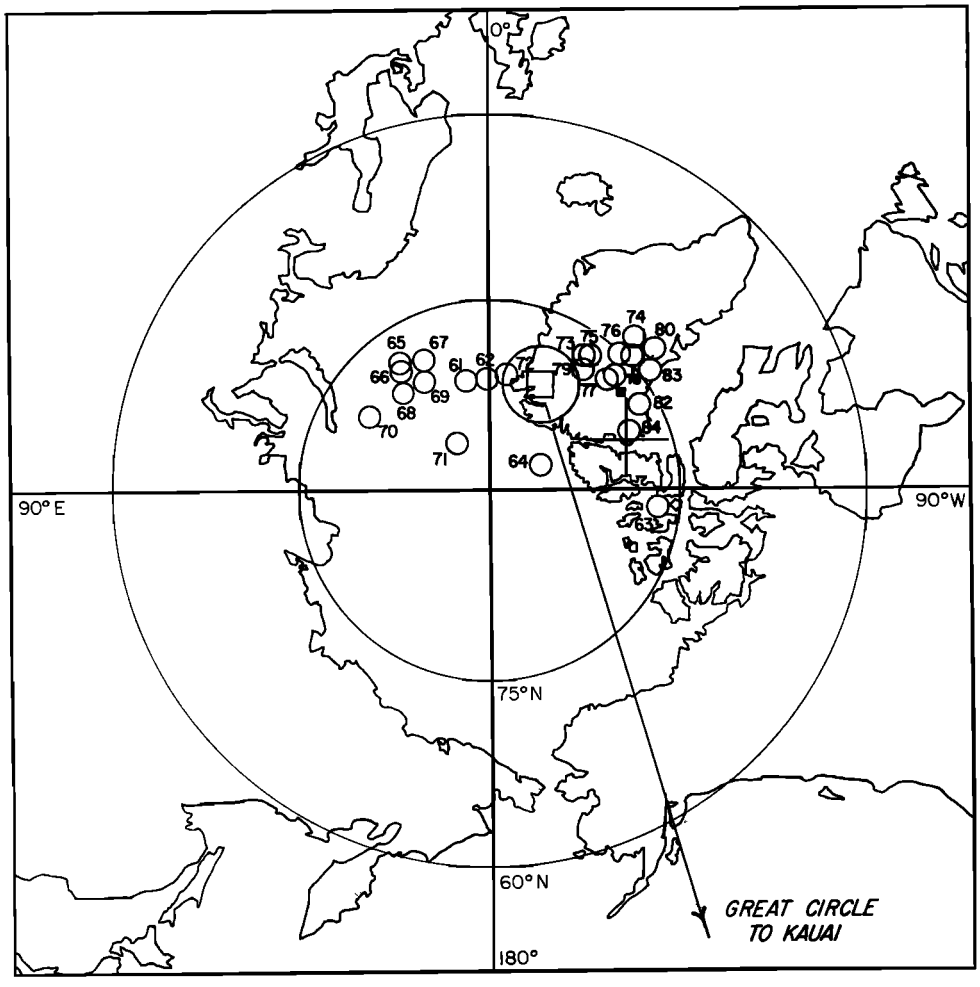


Fig. 5. Virtual geomagnetic pole positions in lavas of the Makaweli formation. (See Figure 4 for explanation.)

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erupting more or less uniformly over the period of 1 m.y., as suggested by *McDougall* [1964]. The magnetic data suggest that the flows accumulated rather rapidly, because they are mostly normally magnetized. Potassium-argon dating of additional lavas in the Napali formation will be necessary before the details of the eruption history are known.

The mean VGP for the Napali formation, Makaweli formation, and Koloa volcanic series are also shown on Figures 4, 5, and 6. Each mean VGP differs significantly from the geographic axis, at the 95% confidence level, if all lavas are treated as statistically independent data. However, as discussed previously [*Doell*, 1969], sequences of superposed lavas such as those at Lawai Gulch, Pueho Ridge (K19–K60), and lavas of the Makaweli formation, rarely satisfy the assumption of not being sequentially related, an assumption that must be satisfied if meaningful confidence limits are to be obtained by *Fisher's* [1953] methods, which were used here.

The three sequences mentioned have been tested for serial correlation through methods developed by *Watson and Beran* [1967]. Briefly, their test is to sum the cosines of the angles between successive directions as observed and compare this sum with similar sums computed from many randomly generated sequences of the same data set. For the Koloa lavas at Lawai Gulch, the observed sum of cosines is

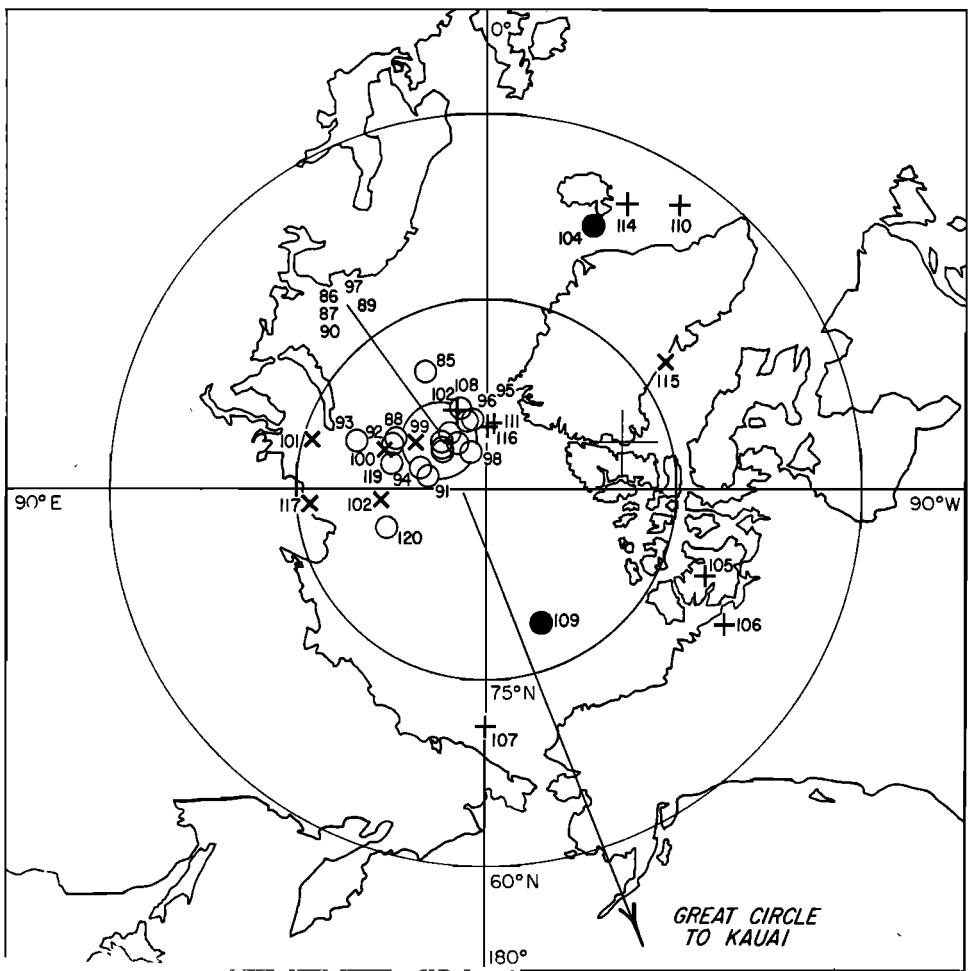


Fig. 6. Virtual geomagnetic pole positions in lavas of the Koloa volcanic series. (See Figure 4 for explanation.)

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TABLE 2. Statistical Analyses of Flow-Average Data

Data Analyzed	Latitude	Longitude	N	R	α_{95}	K_T	S_T
Koloa volcanic series VGP	84.9	42.1	19	18.858	3.0°	126	7.2°
Makaweli formation VGP	80.8	333.7	24	23.750	3.1	92	8.4
Napali formation VGP	82.1	45.7	46	45.038	3.1	47	11.9
All VGP	83.6	23.4	89	87.389	2.0	55	11.0
	Incl.	Decl.	N	R	α_{95}	k_T	s_T
Koloa volcanic series RM directions	31.2	358.1	19	18.769	3.8°	78	9.2°
Makaweli formation RM directions	29.0	6.7	24	23.707	3.4	79	9.2
Napali formation RM directions	26.9	356.5	46	44.398	4.0	28	15.4
All RM directions	28.5	359.6	89	86.637	2.5	37	13.3

N, number of lava flows (sites).
R, vector sum of N unit vectors.
α_{95}, 95% confidence semiangle about means.
K_T, k_T, total precision parameter, VGP and RM directions, respectively.
S_T, s_T, total angular standard deviation about means, VGP and RM directions, respectively.
All reversed flow-average data converted to normal north-seeking data before analysis.

between one and two standard deviations greater than the sum expected if the data were unrelated; for the Napali and Makaweli formation sequences the sums are more than four standard deviations larger than expected. Thus at the 95% confidence level (about two standard deviations) we may reject the hypothesis of serial correlation for the Koloa samples, but not reject it for the others. Except possibly for the Koloa volcanic series, then, the values of α_{95} shown in Figures 4, 5, and 6 and listed in Table 2 are smaller than they should be.

It is interesting to note that the mean VGP for each unit, as well as the over-all mean (Table 2), fall on the side of the geographic pole farthest removed from the sampling sites on Kauai. *Wilson* [1970] has noted this same feature for many worldwide paleomagnetic studies of Tertiary to Holocene rocks. He suggested that this might indicate a permanent displacement of the 'main' geomagnetic axial dipole toward the north pole. The Kauai data tend to support this hypothesis, but because of the statistical uncertainty stated above, it is not clear at what level of significance one can take the difference between the VGP and the geographic axis.

Another possible interpretation for this 'far-side' displacement of the mean VGP is that Kauai was at a significantly lower latitude when these lavas erupted and has since drifted northward. This interpretation is unlikely, however, because the same far-side effect has been noted in the mean VGP for Brunhes age (<700,000 years old) lavas from the islands of Hawaii and Oahu. There is no significant difference in the displacements of the 4.5- to 5.5-m.y.-old Napali formation VGP and 1.4-m.y.-old Koloa volcanic series VGP.

The total angular variances observed in the Kauai paleomagnetic data are given in the last two columns of Table 2 and are further analyzed in Table 3. The methods used to calculate the values cited are based on procedures developed by *Fisher* [1953], *Watson and Irving* [1957], and *Cox* [1969]. The exact manner in which these procedures have been employed is given by *Doell* [1970]. A value of $S_A = 1.25°$ was assumed as the contribution to the total variance due to local magnetic anomalies at the time of cooling of the lavas [*Doell and Cox*, 1963]. This value was used in the conversion from between flow variance (Table 3, subscript B) to ancient geomagnetic variance (subcript F).

The variances have been presented in Table 3 in several different measures for both VGP and directions so that they can be compared with previously determined paleosecular variation data in any desired form. For reasons previously stated [*Doell*, 1970; *Doell and Cox*, 1972], which include the fact that the rotation axis seems to be the long-term symmetry axis for the geomagnetic field [*Cox and Doell*, 1960; *Irving*, 1964], it is preferred here to use the angular standard deviation of VGP measured about the geographic axis (S_F) as the best

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TABLE 3. Dispersion Statistics of Flow-Average Data

Data Analyzed	$\langle N \rangle$	Analyses of VGP								Analyses of RM directions							
		S_T	S_W	S_B	S_F	K_T	K_W	K_B	K_F	s_T	s_W	s_B	s_F	k_T	k_W	k_B	k_F
Referred to means																	
Koloa volcanic series	4.89	7.2°	5.7°	6.8°	6.6°	126	201	155	167	9.2°	4.1°	8.7°	8.6°	78	388	86	88
Makaweli formation	8.21	8.4	5.9	8.2	8.0	92	163	97	102	9.2	4.0	9.0	8.9	79	359	80	82
Napali formation	4.98	11.9	5.0	11.7	11.6	47	241	49	50	15.4	3.6	15.3	15.2	28	454	28	28
All flow-average data	5.83	11.0	5.4	10.8	10.7	55	196	58	60	13.3	3.7	13.2	13.1	37	400	40	40
Referred to axial geocentric dipole																	
Koloa volcanic series	4.89	8.9°	5.7°	8.6°	8.4°	88	201	156	168	12.2°	4.1°	12.1°	12.0°	47	388	54	54
Makaweli formation	8.21	12.6	5.9	12.4	12.3	43	163	59	60	14.8	4.0	14.8	14.7	31	359	34	35
Napali formation	4.98	14.3	5.0	14.1	14.0	33	241	38	39	19.8	3.6	19.7	19.7	17	454	18	18
All flow-average data	5.83	12.7	5.4	12.5	12.4	42	196	53	54	17.0	3.7	16.9	16.9	23	400	25	25

$\langle N \rangle$, mean number of sample cores per lava flow.
S, s, angular standard deviations for VGP and RM directions, respectively.
K, k, precision parameters for VGP and RM directions, respectively.
T, total dispersion.
W, within-flow dispersion.
B, between-flow dispersion.
F, ancient geomagnetic field dispersion.

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measure of paleosecular variation for comparison with similar data from other localities. If *Wilson's* [1970] hypothesis is correct, however, it might be better to use variances calculated about means, or more properly perhaps, variances calculated about the VGP that would be produced from the field in Kauai by the northward displaced dipole that best fits all worldwide paleomagnetic data of comparable ages. As may be seen in comparing data referred to means with data referred to the axial geocentric dipole in Table 3, the effect is to reduce somewhat the paleosecular variation. So long as each area being compared had a mean far side pole, the relative change in paleosecular variation would not be very great. The effect is similar to that of modifying the models for paleosecular variation [*Cox*, 1970] by subtracting a bit more dipole wobble, which produces the same contribution to S_F at all localities.

The values of S_F in Table 3 show that paleosecular variation seems to increase with the age of the Kauai lavas and that the values are relatively small when compared with values for other regions [*Doell and Cox*, 1971, 1972]. However, it is very difficult to place confidence limits on the values of S_F because the data, at least in part, are strongly serially correlated. Formal limits derived by *Cox's* [1969] methods and by assuming all data independent will yield minimum limits; they are 6.9° to 10.8° for the Koloa volcanic series, 10.2° to 15.4° for the Makaweli formation, and 12.2° to 16.4° for the Napali formation. The range of S_F for all data is 11.2° to 13.9°. The true ranges are certainly broader, and possibly no significance should be attached to the increasing values of variance with age. On the other hand, such a relation with age has also been found in France [*Doell*, 1970] and Iceland [*Doell*, 1972].

The over-all low value for paleosecular variation in the Kauai lavas is compatible with the present low secular variation and nondipole strength in the central Pacific [*Doell and Cox*, 1972], the small variation between paleomagnetic directions in historically erupted lavas [*Doell and Cox*, 1963], and the low angular variance given in several paleomagnetic studies of Brunhes age lavas from Hawaii and Oahu [*Doell and Cox*, 1965; *Doell*, 1969, 1972]. It is suggested, therefore, that the present anomalously low secular variation and nondipole field strength has been a permanent feature of the central Pacific Ocean for at least 5 m.y.

APPENDIX: SITE LOCATION NOTES

The quadrangle maps referred to are all 7.5-min series (1:24,000 scale) U.S. Geological Survey topographic maps dated either 1963 or 1965. The latitude and longitude of the sampling site are given in parentheses.

K120: Middle of three flows exposed in quarry 200 meters north of junction of Highway 50 and Kahili Road, NE Koloa quadrangle (21.968°N, 200.548°E).

K119: Lower flow at same location as K120.

K118: Samples from residual 'eggs' in weathered outcrops just SE of Highway 56 bridge over Anahola Stream, SE Anahola quadrangle (22.145°N, 200.685°E).

K117: Lava exposed at sea level about 200 meters SSW of Mokolea Point on west shore of Kilauea Bay, NW Anahola quadrangle (22.226°N, 200.614°E).

K116: Lava flow above K117 about 70 meters toward Mokolea Point.

K115: Samples from residual 'eggs' in ditch north of Kanehu Reservoir 2, 3½ km WSW of Anahola, south center Anahola quadrangle (22.133°N, 200.651°E).

K114: Samples from residual 'eggs' on south side of private road 130 meters east of west edge of Kapaa quadrangle about 1 ¾ km WNW of Kapaia Reservoir, SW Kapaa quadrangle (22.028°N, 200.584°E).

K113: Samples from residual 'eggs' on both sides of private road just south of junction marked 610, SE of Hanahanapuni, east center Waialeale quadrangle (22.051°N, 200.579°E).

K112: Samples in quarry 100 meters north of south fork Wailua River, 100 meters west of east edge of Waialeale quadrangle, east center of quadrangle (22.033°N, 200.582°E).

K111: Flow in river bed 100 meters upstream of Wailua Falls on south fork of Wailua River, west center Kapaa quadrangle (22.038°N, 200.617°E).

K110: Samples from residual 'eggs' in roadcuts on east side of private road about 500 meters SSE of junction 330 just east of Pukaki Reservoir, SW Kapaa quadrangle (22.011°N, 200.617°E).

K85 through K109: Superposed flows in roadcut on west side of private road just west of Lawai Stream crossing, 500 meters east of Luawai Reservoir, SW Koloa quadrangle K109 where road turns sharply to north (21.898°N, 200.492°E). K85 near gaging station on Lawai Stream (21.902°N, 200.495°E).

K61 through K84: Superposed lavas in bottom of stream that flows east into Waimea River about 800 meters south of Poo Kaeha in Waimea Canyon, west center Waimea Canyon quadrangle. K61 200 meters west of Waimea River (22.067°N,

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200.354°E) with flows extending about 500 meters upstream to K84 (22.066°N, 200.348°E).

K19 through K60: Superposed flows sampled in roadcuts on NW side of Highway 55 where it ascends the SE side of Puehu Ridge 4 km north of Kekaha, south center Kekaha quadrangle. K19 at Waipao Valley Stream (22.009°N, 200.297°E); K60 where road turns sharply north on Puehu Ridge (22.291°N, 200.005°E).

K7 through K18: Superposed lava flows exposed along or near Kukui Trail on west side of Waimea Canyon just east of Kukui, west center Waimea Canyon quadrangle. K7 near 600-meter elevation (22.057°N, 200.346°E); K18 near 900-meter elevation (22.045°N, 200.339°E).

K4 through K6: Superposed lava flows in bottom of Waimea Canyon just south of dam on Waimea River 400 meters north of Poo Kaeha, NW Waimea Canyon quadrangle (22.077°N, 200.349°E).

K1 through K3: Superposed lavas just west of Haena Dry Cave 400 meters SW of Haena, NE Haena quadrangle (22.223°N, 200.432°E).

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